

|              |                                          |                                         |                              |
|--------------|------------------------------------------|-----------------------------------------|------------------------------|
| Name         | : Mr. VIPAN KUMAR GARG                   | Age                                     | : 68 Years                   |
| Lab No.      | : 193146741                              | Gender                                  | : Male                       |
| Ref By       | : PGI CHANDIGARH                         | Reported                                | : 10/2/2025 1:12:33PM        |
| Collected    | : 7/2/2025 9:40:00AM                     | Report Status                           | : Final                      |
| A/c Status   | : P                                      | Processed at                            | : LPL-NATIONAL REFERENCE LAB |
| Collected at | : CHANDIGARH-IV                          | National Reference laboratory, Block E, |                              |
|              | Booth Number-122,Sector 10, Inner Market | Sector 18, Rohini, New Delhi -110085    |                              |
|              | Panchkula                                |                                         |                              |
|              | PANCHKULA                                |                                         |                              |



### Test Report

| Test Name | Results | Units | Bio. Ref. Interval |
|-----------|---------|-------|--------------------|
|-----------|---------|-------|--------------------|

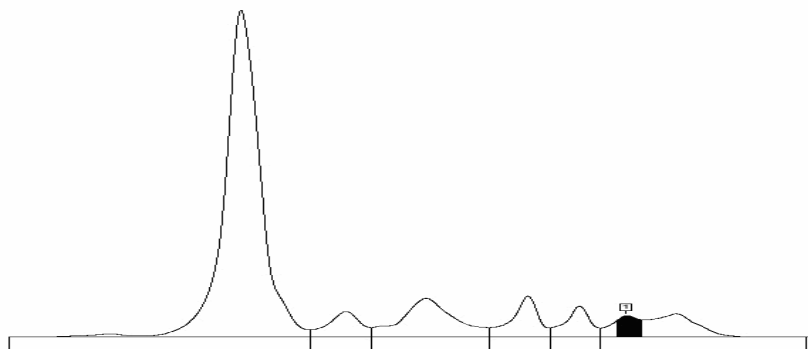
#### MULTIPLE MYELOMA MONITORING PANEL

#### PROTEIN ELECTROPHORESIS, SERUM

(Capillary Electrophoresis)

|                  |             |      |             |
|------------------|-------------|------|-------------|
| Protein, Total   | 7.20        | g/dL | 6.40 - 8.10 |
| Albumin          | 4.69        | g/dL | 3.60 - 5.40 |
| Alpha 1 globulin | 0.32        | g/dL | 0.20 - 0.40 |
| Alpha 2 globulin | 0.80        | g/dL | 0.50 - 1.00 |
| Beta 1 globulin  | <b>0.42</b> | g/dL | 0.50 - 1.10 |
| Beta 2 globulin  | 0.30        | g/dL | 0.30 - 0.60 |
| Gamma globulin   | <b>0.66</b> | g/dL | 0.70 - 1.50 |
| A : G Ratio      | 1.87        |      | 0.90 - 2.00 |
| M Spike          | ? 0.2       | g/dL |             |

ID **193146741**



|              |                                          |               |                                           |
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Interpretation

Suspicious non-discrete protein band in gamma globulin region, possibly Monoclonal gammopathy. Advised: Immunofixation electrophoresis (IFE) to confirm the presence of monoclonal gammopathy and identify its immunological nature.

Comments

Serum Protein electrophoresis (SPE) is used to identify patients with Monoclonal gammopathies (Multiple Myeloma (MM), Waldenstrom's macroglobulinemia, AL Amyloidosis as well as premalignant conditions Monoclonal gammopathy of unknown significance(MGUS) and Smoldering Myeloma) and other serum protein abnormalities (Nephrotic syndrome, Alpha 1 antitrypsin disorder, and inflammatory processes associated with infection, liver diseases & autoimmune disorder). Monoclonal gammopathies indicate clonal expansion of plasma cells or mature B cells. Monoclonal gammopathies are present in 8% of geriatric patients and require further evaluation for monitoring & prognosis. SPE can be used for monitoring response to therapy, a decrease or increase of M spike by 0.5 g/dl is considered significant change.

If SPE alone is used as initial diagnostic screen it will be able to detect approximately 88% of all MM, 66% of AL Amyloidosis and 56% of Light Chain Deposition Disease (LCDD) patients. SPE has good sensitivity in detecting intact monoclonal protein but has limited sensitivity in detecting monoclonal free light chains (FLC). Therefore, SPE alone will miss many patients that are only producing monoclonal FLC. These misses include many Light Chain MM, all Non Secretory MM and significant numbers of AL amyloidosis and LCDD patients. Combination of FLC assay with SPE & immunofixation improves the sensitivity to 99%. Thus, Multiple Myeloma screening panel which includes SPE, IFE and FLC should be done to screen patients with high clinical suspicion for MM.

IMMUNOTYPING, SERUM



|              |                                          |               |                                           |
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| Lab No.      | : 193146741                              | Gender        | : Male                                    |
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|              | Panchkula                                |               |                                           |
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|           |         |       |                    |

**Interpretation:-**Serum protein profile by high-resolution Agarose gel electrophoresis reveals the presence of a discrete protein band in gamma globulin region suggestive of 'M' spike  
A 2nd protein band proximal to it is also seen in gamma globulin region could be isomer of the same.  
Immunofixation electrophoresis (IFE) identifies the 'M' spike as IgG Kappa and its (?) isomer as IgG Kappa .

**Note:** Immunotyping is not a quantitative assay. If a monoclonal protein is identified, a serum protein electrophoresis assay is required for quantifying the abnormality.

Comment

The assay is used for immunotyping of monoclonal proteins which identifies the monoclonal immunoglobulin heavy-chain (gamma, alpha, mu) and/or light-chain type (kappa or lambda). It is generally recommended that both serum Protein electrophoresis (SPEP) and Immunotyping be used as a screening panel because it is more sensitive than SPEP. Immunotyping is not only recommended as part of the initial screening process but also for confirmation of complete response to therapy.

Usage

Identification of monoclonal immunoglobulin heavy and light chains  
Documentation of complete response to therapy



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### Test Report

| Test Name                                    | Results     | Units | Bio. Ref. Interval |
|----------------------------------------------|-------------|-------|--------------------|
| <b>BETA 2 MICROGLOBULIN, SERUM</b><br>(CLIA) |             |       |                    |
| Beta-2 Microglobulin, Serum                  | <b>7864</b> | ng/mL | 609.00 - 2366.00   |

#### Note

- False negative / positive results are observed in patients receiving mouse monoclonal antibodies for diagnosis or therapy.
- Beta 2 Microglobulin values regardless of levels should not be interpreted as absolute evidence of the presence or absence of disease. All values should be correlated with clinical findings and results of other investigations.

#### Comment

As per International Staging System for Multiple Myeloma the level of Beta 2 microglobulin can be used for staging the disease:

| STAGE     | BETA 2 MICROGLOBULIN LEVEL IN (ng/mL) |
|-----------|---------------------------------------|
| Stage I   | <3500                                 |
| Stage II  | 3500-5500                             |
| Stage III | >5500                                 |

#### Increased levels

- \* Lymphoproliferative disorders like Multiple myeloma, B cell Lymphoma and Chronic Lymphocytic Leukemia.
- \* Inflammatory disorders like Rheumatoid arthritis, SLE, Sjogren's syndrome and Crohn's disease.
- \* Renal Dysfunction

#### Uses

- \* Prognostic indicator of Multiple myeloma and other hematopoietic malignancies
- \* An aid in the management of patients with Renal dysfunction and Rheumatoid arthritis
- \* Staging of Multiple Myeloma

#### IMMUNOGLOBULIN PROFILE, SERUM





|              |                                                                                     |               |                              |                                                                                     |
|--------------|-------------------------------------------------------------------------------------|---------------|------------------------------|-------------------------------------------------------------------------------------|
| Name         | : Mr. VIPAN KUMAR GARG                                                              | Age           | : 68 Years                   |  |
| Lab No.      | : 193146741                                                                         | Gender        | : Male                       |                                                                                     |
| Ref By       | : PGI CHANDIGARH                                                                    | Reported      | : 10/2/2025 1:12:33PM        |                                                                                     |
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| A/c Status   | : P                                                                                 | Processed at  | : LPL-NATIONAL REFERENCE LAB |                                                                                     |
| Collected at | CHANDIGARH-IV<br>Booth Number-122,Sector 10, Inner Market<br>Panchkula<br>PANCHKULA |               |                              | National Reference laboratory, Block E,<br>Sector 18, Rohini, New Delhi -110085     |

### Test Report

| Test Name                                        | Results | Units | Bio. Ref. Interval |
|--------------------------------------------------|---------|-------|--------------------|
| Immunoglobulin IgG, Serum<br>Immunoturbidimetry) | 736.00  | mg/dL | 650.00 - 1600.00   |
| Immunoglobulin IgM, Serum                        | <21.0   | mg/dL | 50.00 - 300.00     |
| Immunoglobulin IgA, Serum                        | 52.70   | mg/dL | 40.00 - 350.00     |

#### Comments:

Approximately 80% of serum immunoglobulin is IgG, 6% is IgM and 13% is IgA. High levels of IgM signify acute infections whereas IgG predominates in chronic infections. IgA is the predominant immunoglobulin in body secretions like saliva, sweat and colostrums.

#### Polyclonal increases are seen in:

- IgG: SLE, Chronic liver diseases, Infectious diseases and Cystic fibrosis
- IgM: Viral, bacterial and parasitic infections, Liver diseases, Rheumatoid arthritis, Scleroderma, Cystic fibrosis & heroin addiction
- IgA: Chronic liver diseases, Chronic infections, Autoimmune disorders, Sarcoidosis and Wiscott-Aldrich syndrome.

**Monoclonal increases** are seen in IgG & IgA Myelomas and IgM increases in cases of Waldenstroms macroglobulinemia

**Decreased synthesis** of IgG, IgM & IgA is seen in Congenital and Acquired Immunodeficiency diseases.

**Decreased levels** are seen in Protein losing enteropathies, Nephrotic syndrome and skin burns.



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Booth Number-122, Sector 10, Inner Market  
Panchkula  
PANCHKULA

Age : 68 Years

Gender : Male

Reported : 10/2/2025 1:12:33PM

Report Status : Final

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National Reference laboratory, Block E,  
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### Test Report

| Test Name                                                               | Results      | Units | Bio. Ref. Interval |
|-------------------------------------------------------------------------|--------------|-------|--------------------|
| <b>KAPPA / LAMBDA LIGHT CHAINS, FREE, SERUM</b><br>(Immunoturbidimetry) |              |       |                    |
| Kappa, Free Light Chain                                                 | <b>46.57</b> | mg/L  | 3.30 - 19.40       |
| Lambda, Free Light Chain                                                | 15.04        | mg/L  | 5.71 - 26.30       |
| Kappa/Lambda(FLC), Ratio                                                | <b>3.096</b> |       | 0.26 - 1.65        |

### Interpretation

| KAPPA  | LAMBDA | KAPPA / LAMBDA RATIO | REMARKS                                                                                                                                                                            |
|--------|--------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normal | Normal | Normal               | Normal sample. If serum electrophoretic pattern is normal, it is unlikely that patient has Monoclonal gammopathy.                                                                  |
| Normal | Normal | Low/High             | Suggestive of Monoclonal gammopathy with Bone Marrow suppression                                                                                                                   |
| Low    | Low    | Normal               | Suggestive of Bone Marrow suppression                                                                                                                                              |
| Low    | Low    | Low/High             | Suggestive of Monoclonal gammopathy with Bone Marrow suppression                                                                                                                   |
| High   | High   | Normal               | Suggestive of :<br>• Renal Impairment<br>• Overproduction of Polyclonal free light chain from inflammatory conditions<br>• Biclinal gammopathy of different free light chain types |
| High   | High   | Low/High             | Suggestive of Monoclonal gammopathy with Renal Impairment                                                                                                                          |

**Note:** International Myeloma Working Group (IMWG-2014) has updated the criteria for the diagnosis of Multiple Myeloma (MM) which includes clonal bone marrow plasma cells ( $\geq 10\%$ ) or biopsy proven Plasmacytoma, with one or more 'Myeloma defining event'. An elevated involved : uninvolved serum FLC ratio  $>100$  (and involved FLC  $\geq 100\text{mg/L}$  is now considered a 'Myeloma defining event'.

### Comment

Measurement of free light chain (FLCs) aids in the diagnosis & monitoring of Multiple myeloma, Waldenstrom's macroglobulinemia, AL Amyloidosis & Light chain deposition disease. Serum concentration of FLCs is dependent upon the balance between production by plasma cells & their progenitors and renal clearance.



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## Test Report

| Test Name                                                                                                                                                                                                                                                                                                                                                                                               | Results | Units | Bio. Ref. Interval |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|--------------------|
| When there is increased polyclonal immunoglobulin production and/or renal impairment, concentration of both kappa & lambda light chains can increase by 30-40 fold but the Kappa / Lambda ratio remains unchanged. In contrast, tumor produces monoclonal excess of only 1 type of light chains, often with bone marrow suppression of alternate light chains resulting in abnormal kappa/lambda ratio. |         |       |                    |

The combination of Serum protein electrophoresis (SPE) & Immunofixation (IFE) are highly sensitive but fail to identify a few patients with Light Chain Multiple Myeloma & all patients of Non Secretory Multiple Myeloma. Combination of free light chain assay with SPE & IFE improves the sensitivity to 99%.

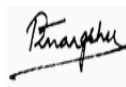
## Uses

### As per recommendations by IMWG:

- FLC,SPE and serum IFE for screening
- For diagnosis and prognosis of all patients with MGUS, Smoldering or active multiple myeloma, Solitary plasmacytoma and AL amyloidosis
- Serial measurements routinely in patients with AL amyloidosis and Multiple myeloma
- patients with oligosecretory disease
- In all patients who have achieved a complete response (CR) to assess for stringent CR

  
DMC - 67327

Dr Anjalika Goyal  
MD,Biochemistry  
Consultant Biochemist  
NRL - Dr Lal PathLabs Ltd

  
DMC - 89819

Dr Himangshu Mazumdar  
MD, Biochemistry  
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Dr Nimmi Kansal  
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Chemistry & Biochemical Genetics  
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Dr.Richa Sirohi  
MD, Biochemistry  
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NRL - Dr Lal PathLabs Ltd



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PANCHKULA

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Plot No. 17, Ground Floor, Industrial Area,  
Phase-1, Backside Citroen Showroom,  
Chandigarh-160002.



### Test Report

| Test Name                                                    | Results | Units         | Bio. Ref. Interval |
|--------------------------------------------------------------|---------|---------------|--------------------|
| <b>MULTIPLE MYELOMA MONITORING PANEL</b>                     |         |               |                    |
| <b>ALBUMIN, SERUM</b><br>(BCG)                               | 4.42    | g/dL          | 3.20 - 4.60        |
| <b>CREATININE, SERUM</b><br>(Modified Jaffe, IDMS traceable) |         |               |                    |
| Creatinine                                                   | 1.82    | mg/dL         | 0.70 - 1.30        |
| GFR Estimated                                                | 40      | mL/min/1.73m2 | >59                |
| GFR Category                                                 | G3b     |               |                    |

#### Advise

1. CKD Risk Map (Z1014)
2. Cystatin C, serum (B173)

#### Note

1. GFR, estimated (eGFR) calculated using the 2021 CKD-EPI creatinine equation and GFR Category reported as per KDIGO guideline 2012.
2. eGFR category G1 or G2 does not fulfil the criteria for CKD, in the absence of evidence of kidney damage

|                                         |      |       |              |
|-----------------------------------------|------|-------|--------------|
| <b>CALCIUM, SERUM</b><br>(Arsenazo III) | 9.60 | mg/dL | 8.80 - 10.20 |
|-----------------------------------------|------|-------|--------------|

### MULTIPLE MYELOMA MONITORING PANEL

|                                                   |        |     |           |
|---------------------------------------------------|--------|-----|-----------|
| <b>LDH;LACTATE DEHYDROGENASE, SERUM</b><br>(IFCC) | 201.00 | U/L | 120 - 246 |
|---------------------------------------------------|--------|-----|-----------|

#### Comments

Lactate dehydrogenase (LDH) is a nonspecific enzyme found in most organs. Highest concentrations are found in liver, heart, kidney and blood cells. LDH measurements are used in the diagnosis and treatments of liver diseases like Acute viral hepatitis, Cirrhosis & Metastatic carcinoma; Cardiac diseases like Myocardial infarction; Tumors of lungs / kidneys & Hematologic disorders like Megaloblastic anemia & Hemolytic anemia.





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Test Report

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|------------------------------|---------|-------|--------------------|
| UREA, SERUM<br>(Urease GLDH) |         |       |                    |
| Urea                         | 38.00   | mg/dL | 17.00 - 49.00      |
| BUN; Blood Urea Nitrogen     | 17.75   | mg/dL | 8.00 - 23.00       |



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### Test Report

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|----------------------------------------------------------------------------------------------------|---------|----------|--------------------|
| <b>COMPLETE BLOOD COUNT;CBC</b><br>(Photometry, electrical impedance, vcs technology & calculated) |         |          |                    |
| Hemoglobin                                                                                         | 11.90   | g/dL     | 13.00 - 17.00      |
| Packed Cell Volume (PCV)                                                                           | 35.40   | %        | 40.00 - 50.00      |
| RBC Count                                                                                          | 3.90    | mill/mm3 | 4.50 - 5.50        |
| MCV                                                                                                | 91.00   | fL       | 83.00 - 101.00     |
| Mentzer Index                                                                                      | 23.3    |          |                    |
| MCH                                                                                                | 30.40   | pg       | 27.00 - 32.00      |
| MCHC                                                                                               | 33.50   | g/dL     | 31.50 - 34.50      |
| Red Cell Distribution Width (RDW)                                                                  | 14.70   | %        | 11.60 - 14.00      |
| Total Leukocyte Count (TLC)                                                                        | 4.30    | thou/mm3 | 4.00 - 10.00       |
| <b>Differential Leucocyte Count (DLC)</b>                                                          |         |          |                    |
| Segmented Neutrophils                                                                              | 40.30   | %        | 40.00 - 80.00      |
| Lymphocytes                                                                                        | 28.00   | %        | 20.00 - 40.00      |
| Monocytes                                                                                          | 27.30   | %        | 2.00 - 10.00       |
| Eosinophils                                                                                        | 4.00    | %        | 1.00 - 6.00        |
| Basophils                                                                                          | 0.40    | %        | <2.00              |
| <b>Absolute Leucocyte Count</b>                                                                    |         |          |                    |
| Neutrophils                                                                                        | 1.73    | thou/mm3 | 2.00 - 7.00        |
| Lymphocytes                                                                                        | 1.20    | thou/mm3 | 1.00 - 3.00        |
| Monocytes                                                                                          | 1.17    | thou/mm3 | 0.20 - 1.00        |
| Eosinophils                                                                                        | 0.17    | thou/mm3 | 0.02 - 0.50        |



|                       |                                                  |                                                    |                            |
|-----------------------|--------------------------------------------------|----------------------------------------------------|----------------------------|
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| <b>Lab No.</b> :      | <b>193146741</b>                                 | <b>Gender</b> :                                    | <b>Male</b>                |
| <b>Ref By</b> :       | <b>PGI CHANDIGARH</b>                            | <b>Reported</b> :                                  | <b>10/2/2025 1:12:33PM</b> |
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| <b>Collected at</b> : | <b>CHANDIGARH-IV</b>                             | <b>Plot No. 17, Ground Floor, Industrial Area,</b> |                            |
|                       | <b>Booth Number-122, Sector 10, Inner Market</b> | <b>Phase-1, Backside Citroen Showroom,</b>         |                            |
|                       | <b>Panchkula</b>                                 | <b>Chandigarh-160002.</b>                          |                            |
|                       | <b>PANCHKULA</b>                                 |                                                    |                            |



### Test Report

| Test Name                                                 | Results | Units    | Bio. Ref. Interval |
|-----------------------------------------------------------|---------|----------|--------------------|
| Basophils                                                 | 0.02    | thou/mm3 | 0.02 - 0.10        |
| Platelet Count                                            | 167     | thou/mm3 | 150.00 - 410.00    |
| Mean Platelet Volume                                      | 9.7     | fL       | 6.5 - 12.0         |
| There is monocytosis<br>Followup and clinical correlation |         |          |                    |

### Comment

In anaemic conditions Mentzer index is used to differentiate Iron Deficiency Anaemia from Beta- Thalassemia trait. If Mentzer Index value is >13, there is probability of Iron Deficiency Anaemia. A value <13 indicates likelihood of Beta- Thalassemia trait and Hb HPLC is advised to rule out the Thalassemia trait.

### Note

- As per the recommendation of International council for Standardization in Hematology, the differential leucocyte counts are additionally being reported as absolute numbers of each cell in per unit volume of blood
- Test conducted on EDTA whole blood



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| <b>LIVER PANEL 1; LFT,SERUM</b><br>(Spectrophotometry) |         |       |                    |
| AST (SGOT)                                             | 22.0    | U/L   | 19.00 - 48.00      |
| ALT (SGPT)                                             | 12.0    | U/L   | 10.00 - 49.00      |
| AST:ALT Ratio                                          | 1.83    |       | <1.00              |
| GGTP                                                   | 12.0    | U/L   | 0 - 73             |
| Alkaline Phosphatase (ALP)                             | 97.00   | U/L   | 30.00 - 120.00     |
| Bilirubin Total                                        | 0.41    | mg/dL | 0.20 - 1.10        |
| Bilirubin Direct                                       | 0.10    | mg/dL | <0.3               |
| Bilirubin Indirect                                     | 0.31    | mg/dL | <1.10              |
| Total Protein                                          | 6.80    | g/dL  | 5.70 - 8.20        |
| Albumin                                                | 4.42    | g/dL  | 3.20 - 4.60        |
| Globulin(Calculated)                                   | 2.38    | gm/dL | 2.0 - 3.5          |
| A : G Ratio                                            | 1.86    |       | 0.90 - 2.00        |

### Note

1. In an asymptomatic patient, Non alcoholic fatty liver disease (NAFLD) is the most common cause of increased AST, ALT levels. NAFLD is considered as hepatic manifestation of metabolic syndrome.
2. In most type of liver disease, ALT activity is higher than that of AST; exception may be seen in Alcoholic Hepatitis, Hepatic Cirrhosis, and Liver neoplasia. In a patient with Chronic liver disease, AST:ALT ratio>1 is highly suggestive of advanced liver fibrosis.
3. In known cases of Chronic Liver disease due to Viral Hepatitis B & C, Alcoholic liver disease or NAFLD, Enhanced liver fibrosis (ELF) test may be used to evaluate liver fibrosis.
4. In a patient with Chronic Liver disease, AFP and Des-gamma carboxyprothrombin (DCP)/PIVKA II can be used to assess risk for development of Hepatocellular Carcinoma.

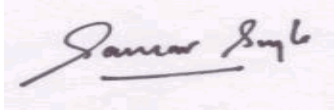
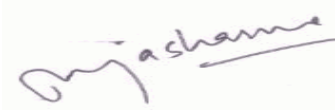




|              |                                          |               |                                             |
|--------------|------------------------------------------|---------------|---------------------------------------------|
| Name         | : Mr. VIPAN KUMAR GARG                   | Age           | : 68 Years                                  |
| Lab No.      | : 193146741                              | Gender        | : Male                                      |
| Ref By       | : PGI CHANDIGARH                         | Reported      | : 10/2/2025 1:12:33PM                       |
| Collected    | : 7/2/2025 9:40:00AM                     | Report Status | : Final                                     |
| A/c Status   | : P                                      | Processed at  | : LPL-CHANDIGARH LAB                        |
| Collected at | : CHANDIGARH-IV                          |               |                                             |
|              | Booth Number-122,Sector 10, Inner Market |               | Plot No. 17, Ground Floor, Industrial Area, |
|              | Panchkula                                |               | Phase-1, Backside Citroen Showroom,         |
|              | PANCHKULA                                |               | Chandigarh-160002.                          |



Test Report

| Test Name                                                                         | Results                                                                           | Units | Bio. Ref. Interval |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------|--------------------|
|  |  |       |                    |
| Dr. Gaurav Singla<br>MBBS MD-Pathology<br>Consultant Pathologist                  | Dr Pooja Sharma<br>MD, Pathology<br>Chief of Laboratory<br>Dr Lal PathLabs Ltd    |       |                    |

-----End of report -----



IMPORTANT INSTRUCTIONS

•Test results released pertain to the specimen submitted. •All test results are dependent on the quality of the sample received by the Laboratory. •Laboratory investigations are only a tool to facilitate in arriving at a diagnosis and should be clinically correlated by the Referring Physician. •Report delivery may be delayed due to unforeseen circumstances. Inconvenience is regretted. •Certain tests may require further testing at additional cost for derivation of exact value. Kindly submit request within 72 hours post reporting. •Test results may show interlaboratory variations. •The Courts/Forum at Delhi shall have exclusive jurisdiction in all disputes/claims concerning the test(s) & or results of test(s). •Test results are not valid for medico legal purposes. •This is computer generated medical diagnostic report that has been validated by Authorized Medical Practitioner/Doctor. •The report does not need physical signature.  
(#) Sample drawn from outside source.  
If Test results are alarming or unexpected, client is advised to contact the Customer Care immediately for possible remedial action.  
Tel: +91-11-49885050,Fax: - +91-11-2788-2134, E-mail: lalpathlabs@lalpathlabs.com

